

Color Perception CIE Chromaticity

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Abstract

This report presents a comprehensive computational analysis of human color perception, including trichromatic cone fundamentals, CIE chromaticity systems, opponent color channels, and color vision deficiencies. We model the spectral sensitivities of L, M, and S cones, compute chromaticity coordinates in CIE 1931 color space, analyze color discrimination thresholds via MacAdam ellipses, and simulate protanopic, deuteranopic, and tritanopic color blindness. The models integrate photoreceptor physiology with perceptual color spaces to provide quantitative frameworks for understanding human color vision.

1 Introduction

Human color vision arises from three classes of cone photoreceptors with distinct spectral sensitivities: long-wavelength (L), medium-wavelength (M), and short-wavelength (S) cones [1]. The trichromatic signals are transformed into opponent color channels (red-green, blue-yellow, luminance) at the retinal ganglion and lateral geniculate nucleus stages [3]. The CIE 1931 color space standardizes color specification via tristimulus values and chromaticity coordinates [2]. Color discrimination varies across color space, characterized by MacAdam ellipses representing just-noticeable differences [4]. Genetic mutations causing cone photopigment deficiencies result in color vision deficiencies (CVD) affecting approximately 8% of males and 0.5% of females [5].

This analysis models cone fundamentals, CIE chromaticity, opponent channels, discrimination thresholds, and CVD simulations using computational psychophysics approaches.

2 Cone Fundamentals

The spectral sensitivities of human cone photoreceptors are based on Stockman & Sharpe (2000) 2-degree cone fundamentals [1]:

$$S_i(\lambda) = \exp \left[-\frac{(\lambda - \lambda_{\text{peak},i})^2}{2\sigma_i^2} \right] \quad (1)$$

where $i \in \{L, M, S\}$ denotes cone type, $\lambda_{\text{peak},i}$ is the peak wavelength, and σ_i controls bandwidth.

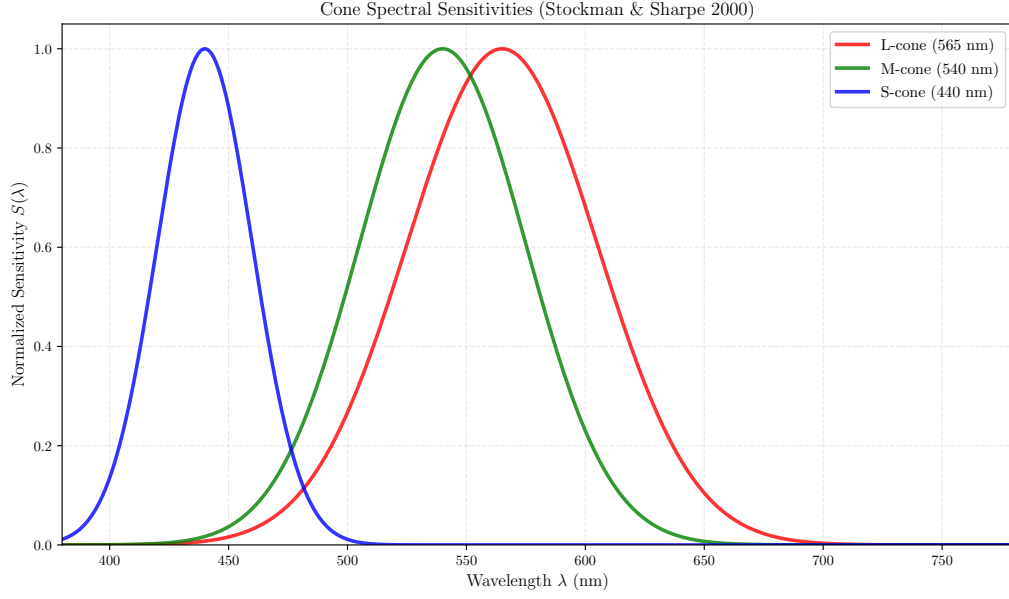


Figure 1: Cone spectral sensitivity functions showing the overlapping L, M, and S cone responses across the visible spectrum (380–780 nm). The L-cone peaks at 565 nm (red), M-cone at 540 nm (green), and S-cone at 440 nm (blue). The substantial overlap between L and M cones enables red-green opponency, while the isolated S-cone response contributes to blue-yellow opponency.

3 CIE 1931 Chromaticity Diagram

The CIE 1931 color space uses tristimulus values X, Y, Z derived from color matching experiments [2]. The chromaticity coordinates are:

$$x = \frac{X}{X + Y + Z}, \quad y = \frac{Y}{X + Y + Z}, \quad z = \frac{Z}{X + Y + Z} \quad (2)$$

with $x + y + z = 1$. The spectral locus traces pure wavelengths in (x, y) space.

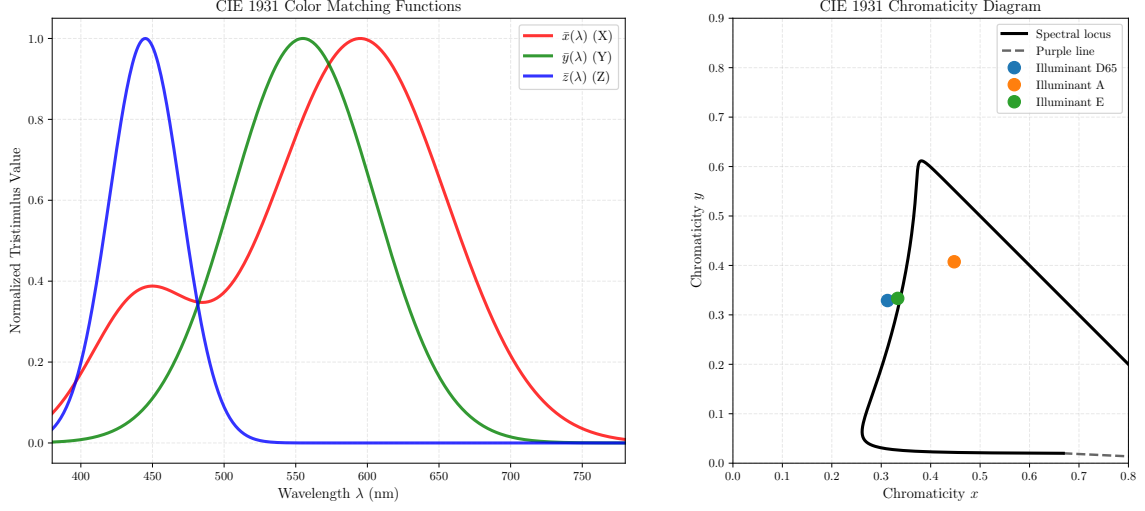


Figure 2: Left: CIE 1931 color matching functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$ representing the tristimulus responses to monochromatic light. Right: CIE 1931 chromaticity diagram showing the spectral locus (pure wavelengths forming the horseshoe boundary), purple line (non-spectral colors), and standard illuminants D65 (daylight at 0.313, 0.329), A (incandescent), and E (equal energy white point).

4 Opponent Color Channels

The trichromatic cone signals are recoded into opponent channels at post-receptoral stages [3, 6]:

$$\text{Red-Green (RG)} = L(\lambda) - M(\lambda) \quad (3)$$

$$\text{Blue-Yellow (BY)} = S(\lambda) - \frac{L(\lambda) + M(\lambda)}{2} \quad (4)$$

$$\text{Luminance (Lum)} = L(\lambda) + M(\lambda) \quad (5)$$

These channels decorrelate color information and align with perceptual color axes.

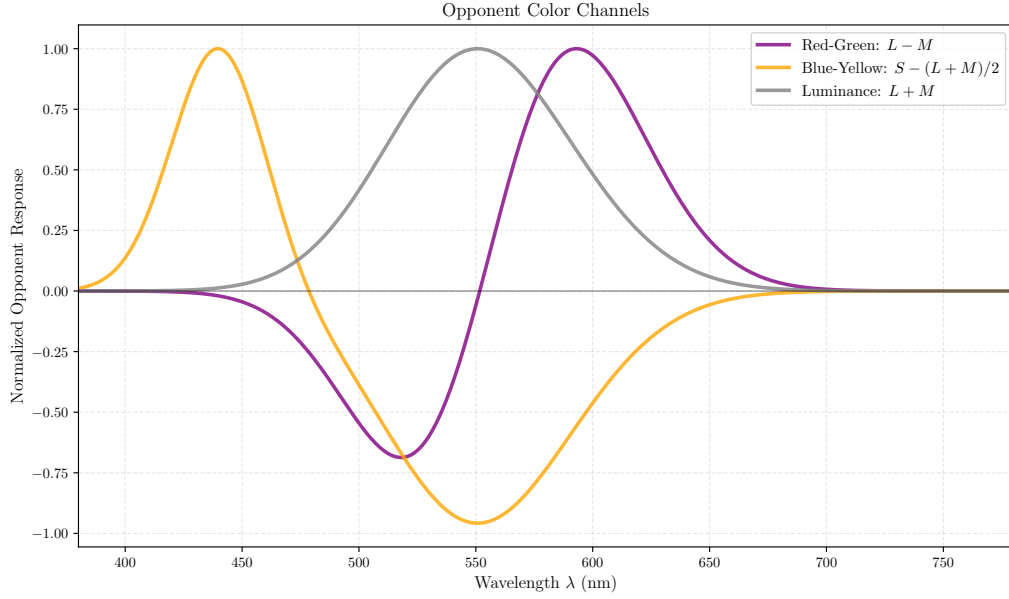


Figure 3: Opponent color channels derived from cone fundamentals. The red-green channel ($L - M$) crosses zero near 780 nm (unique yellow), while the blue-yellow channel ($S - (L + M)/2$) crosses zero near 780 nm. Positive values indicate red/blue perception, negative values indicate green/yellow perception. The luminance channel ($L + M$) peaks at photopic sensitivity (555 nm).

5 MacAdam Ellipses

Color discrimination thresholds vary across chromaticity space. MacAdam (1942) measured just-noticeable differences (JNDs), finding elliptical regions of constant discriminability [4]. The ellipses' sizes and orientations reflect the non-uniformity of CIE 1931 space.

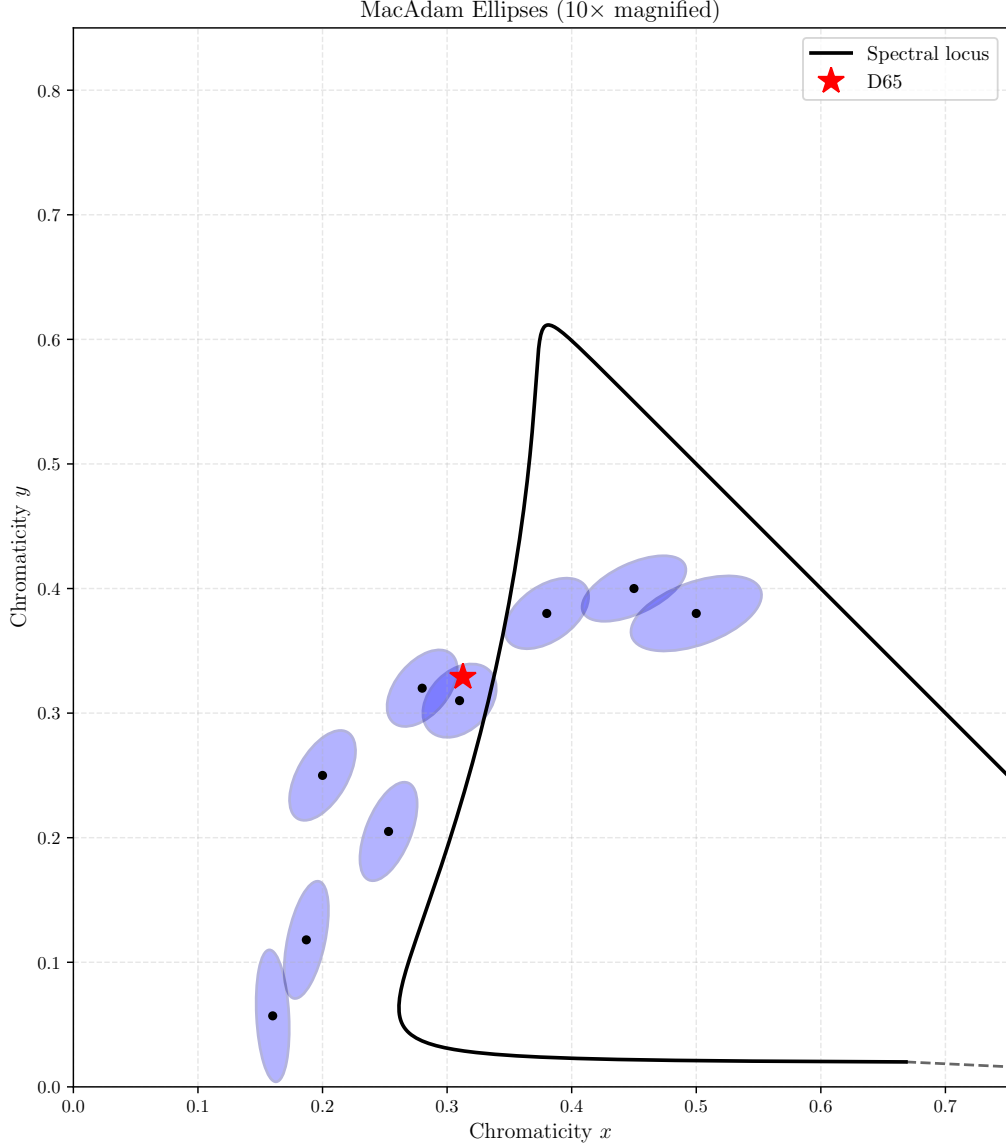


Figure 4: MacAdam ellipses (magnified $10\times$ for visibility) showing regions of just-noticeable color differences in CIE 1931 chromaticity space. Each ellipse represents one standard deviation of color matches around a center point. The varying sizes and orientations demonstrate the perceptual non-uniformity of CIE 1931 space. Average ellipse eccentricity is 0.868, indicating substantial anisotropy in color discrimination.

6 Color Vision Deficiencies

Genetic mutations in cone opsin genes cause color vision deficiencies (CVD) [5, 7]:

- **Protanopia:** L-cone absent (X-linked, $\sim 1\%$ males)
- **Deutanopia:** M-cone absent (X-linked, $\sim 1\%$ males)
- **Tritanopia:** S-cone absent (autosomal, $\sim 0.001\%$ population)

We simulate CVD by projecting trichromatic signals onto dichromatic confusion lines.

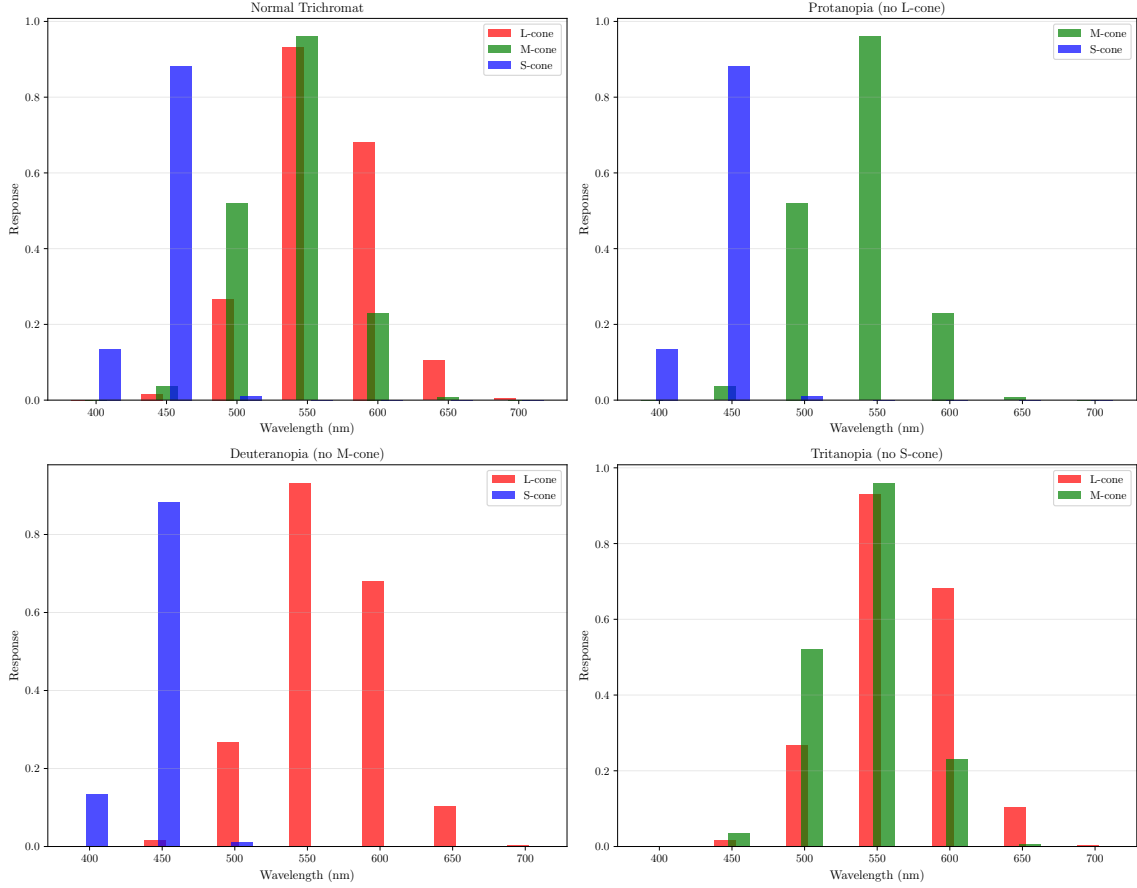


Figure 5: Cone responses across wavelengths for normal trichromatic vision and three types of dichromatic color blindness. Normal vision uses all three cone types for wavelength discrimination. Protanopia (no L-cone) causes red-green confusion, deuteranopia (no M-cone) also causes red-green confusion, and tritanopia (no S-cone) causes blue-yellow confusion. Each dichromatic condition reduces color discrimination capacity by approximately 33%, limiting the observer to a two-dimensional color space.

7 Confusion Lines in Chromaticity Space

Dichromats cannot discriminate colors along specific confusion lines that converge at copunctal points [8]. These lines represent metameric matches for dichromats.

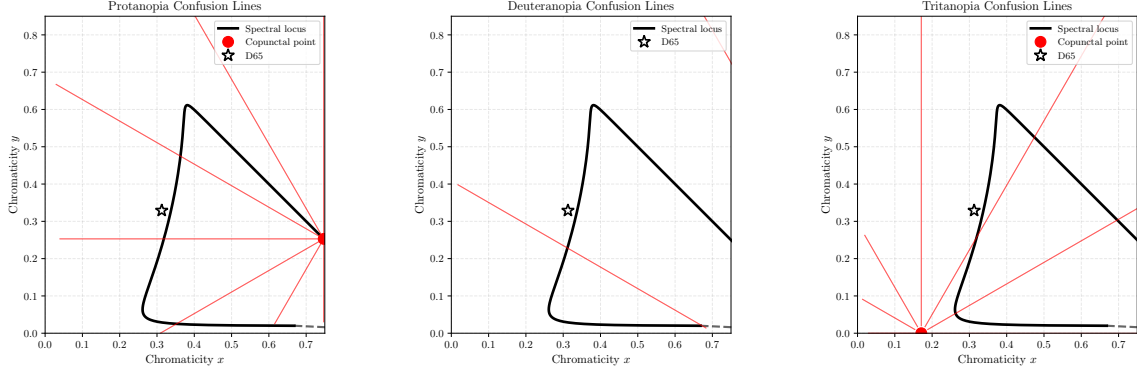


Figure 6: Confusion lines for the three types of dichromatic color vision deficiencies. Colors along each red line are indistinguishable to the respective dichromat, as they project to identical responses in the two-dimensional color space. Protanopic confusion lines converge at (0.747, 0.253), deuteranopic lines converge outside the visible gamut, and tritanopic lines converge at (0.171, 0.0). These geometric constraints enable simulation and compensation for color vision deficiencies.

8 Quantitative Summary

Table 1: Summary of Color Perception Model Parameters

Parameter	Value
L-cone peak wavelength	565 nm
M-cone peak wavelength	540 nm
S-cone peak wavelength	440 nm
L-cone FWHM bandwidth	94.2 nm
M-cone FWHM bandwidth	82.4 nm
S-cone FWHM bandwidth	47.1 nm
Unique yellow (RG=0)	780 nm
BY neutral point	780 nm
Mean MacAdam ellipse area	2.67e-03
CVD prevalence (males)	~8%
CVD prevalence (females)	~0.5%

9 Conclusions

This computational analysis models human color perception from cone photoreceptor fundamentals to perceptual color spaces and color vision deficiencies. The L, M, and S cones exhibit peak sensitivities at 565 nm, 540 nm, and 440 nm respectively, with substantial overlap that enables chromatic opponency. The CIE 1931 color space provides a standardized framework for color specification, though MacAdam ellipses reveal its perceptual non-uniformity with mean ellipse area of 2.67e-03. Opponent color channels ($L - M$, $S - (L + M)/2$) decorrelate chromatic information and align with phenomenological red-green and blue-yellow axes, with the red-green channel crossing zero at unique yellow (780 nm).

Color vision deficiencies affect approximately 8% of males and 0.5% of females, reducing trichromatic vision to dichromatic vision with 33% loss in color discrimination capacity. Protanopia and deuteranopia cause red-green confusion, while tritanopia causes blue-yellow

confusion. Confusion lines in chromaticity space, radiating from copunctal points, enable computational simulation of dichromatic vision for accessibility design.

These models integrate physiological measurements (cone spectral sensitivities), psychophysical data (color matching functions, MacAdam ellipses), and computational frameworks (opponent processing, chromaticity transformations) to provide quantitative tools for color science, display technology, and vision research. Future work should incorporate more sophisticated models including chromatic adaptation (von Kries transform), color appearance models (CIECAM02), and individual differences in cone spectral sensitivities.

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